

Table 1: The different types of databases

Database type	Application and product name
Relational databases (RDBMS)	Storage of data in tables and execution using SQL queries <i>Databases:</i> Oracle, MySQL, SQL Server, PostgreSQL, SQLite
Multi-model databases	Support for assorted models (graph + document) <i>Databases:</i> OrientDB, ArangoDB
Object-oriented databases	Storage of objects with object-oriented programming (OOP) <i>Databases:</i> ObjectDB, db4o
Time-series databases	Optimisation and timestamped data <i>Databases:</i> TimescaleDB, InfluxDB

Table 2: NoSQL databases

Document stores	JSON format documents <i>Databases:</i> CouchDB, MongoDB
Key-value stores	Key-value based <i>Databases:</i> DynamoDB, Redis
Wide-column stores	Column format and for large-scale storage <i>Databases:</i> HBase, Cassandra
Graph databases	Connected datasets with nodes and relations <i>Databases:</i> ArangoDB, Neo4j

applications, and games, which must be integrated with a suitable database application for effective results and decision making.

DuckDB: The open source high-performance database

DuckDB is a powerful and high-performance in-process database for fast analytical processing. It is useful for data science and engineering-based applications and provides a full-featured database without the deployment of a separate dedicated database server.

It can interface with multiple programming languages and scripts including SQL, Python, R, Java, and Node.js. Key features of DuckDB are:

- Cross-platform and lightweight
- Columnar storage and execution
- Native Python and R integration
- Built-in analytical functions
- Direct file access
- Vectorized execution engine
- ACID compliance
- Zero configuration
- Full ANSI SQL support
- Extensible and embeddable
- Export to different formats including Parquet/comma

Table 3: Deployment-based databases

Type	Application and product name
Cloud databases	Cloud deployments and hosted <i>Databases:</i> Amazon RDS, Firebase, Google BigQuery
Embedded databases	Execution with application without dedicated server <i>Databases:</i> SQLite, H2, DuckDB
In-memory databases	RAM based databases <i>Databases:</i> Redis, SAP HANA, Memcached
Distributed databases	Distributed and deployed in assorted servers, nodes and data centers <i>Databases:</i> Google Bigtable, Cassandra
Client-server databases	Dedicated databases for client-server applications <i>Databases:</i> PostgreSQL, MySQL

Table 4: Purpose-oriented databases

Type	Application and product name
Mobile databases	For smartphone and lightweight devices <i>Databases:</i> SQLite, Realm
Analytical databases (OLAP)	For data aggregations and queries <i>Databases:</i> Redshift, Snowflake, DuckDB
Data warehouses	Centralised and high-performance data for real-time analytics <i>Databases:</i> BigQuery, Snowflake, Amazon Redshift
Search databases	Developed and deployed for search queries and full-text analytics <i>Databases:</i> Solr, Elasticsearch
Data lakes	Storage of raw, structured, unstructured data <i>Databases:</i> Apache Iceberg, Delta Lake
Transactional databases (OLTP)	Execution of high-speed rapid transactions <i>Databases:</i> SQL Server, MySQL

- separated value (CSV)
- Works with large data
- Time travel queries

Installing DuckDB and working with Python

DuckDB can be installed with existing Python using the pip installer:

```
$ pip install duckdb
```